

SECTION 2

HIGH PRESSURE AND/OR HELIUM LOSS TROUBLESHOOTING

2-1 INTRODUCTION

Helium loss on a “Zero Boil-Off” magnet should be insignificant (i.e., not measurable) over a period of weeks or months. If significant helium loss is suspected, the following is required to define and quantify the suspected loss:

- The magnet must be in a normal operating mode and have stabilized after a ramp or fill.
- Two sets of data must be taken over time (2 days to 2 weeks, depending on size of leak), with known and calibrated instrumentation and at the same vessel pressure, to establish the extent of the problem.

If high pressure or helium loss is established, use this Section’s troubleshooting guides to determine the root cause(s) of the problem and to resolve the issue(s).

2-2 TOOLS / EQUIPMENT

Make sure the following are available at the site before starting this procedure.

2-2-1 Troubleshooting Tools / Equipment

- Portable Liquid Helium Level Meter (2270571)
- Leak detector fluid or instrument, such as:
 - Bird Dog Quart Spray (p/n DOG-1Q)
 - Big Blue (p/n Rt100S / Rt100G)
 - Snoop
 - Portable Leak Detector
- Leybold Leak Detector
- Portable RuO Temperature Monitor (2171219) (included in RuO Temperature Monitor Kit 2183710)
- Plexiglass Heat Exchanger Cover Plate (2174214)
- Helium gas tubing
- O-Rings included in Spare Parts Kit (46-294744G5)
- Transition Flange O-Ring (46-281247P1)

2-2-1 Troubleshooting Tools / Equipment (continued)

- Cryocooler Maintenance Kit (46-281088G3)
- Field Vacuum Pump Kit (46-294047G1)
- M10 Allen Wrench (46-320470P10)
- Small balloon

2-2-2 De-Icing Tools / Equipment

Required for Shim Lead Connector and Vertical Penetration De-Icing (REPLACEMENT / MAINTENANCE, Section 2-2):

- Lexan Cover Plate (46-294765G1) for Vertical Penetration
- Helium Pressure Regulator and Hose Kit (46-306734G1)
- Nonmagnetic cylinder(s) of helium gas
- Tygon tubing
- Deicing Purge Tube (2347725)
- 1/2 inch (12.7 mm) ice-coring saw (2280035) and 3/4 inch (19.0 mm) ice-coring saw (2278972)
- Flashlight

Required for Recondenser De-Icing (REPLACEMENT / MAINTENANCE, Section 2-4):

- Universal Coldhead Replacement Tool Kit (2300306)
- Two nonmagnetic 235 SCF cylinders of dry nitrogen gas
- Two nonmagnetic 235 SCF cylinders of helium gas (99.9995% pure), each with a low pressure regulator
- Portable RuO Temperature Monitor (2171219) (included in RuO Temperature Monitor Kit 2183710)
- Copper tubing coil
- Heat gun

2-2-3 Coldhead Replacement Tools / Equipment

Required for Two-Stage Cryocooler Coldhead Replacement (REPLACEMENT / MAINTENANCE, Section 12):

- New / Reconditioned Coldhead* (2218465)

Note

New / reconditioned Coldhead must have been inspected and be free of defects before bringing to the site.

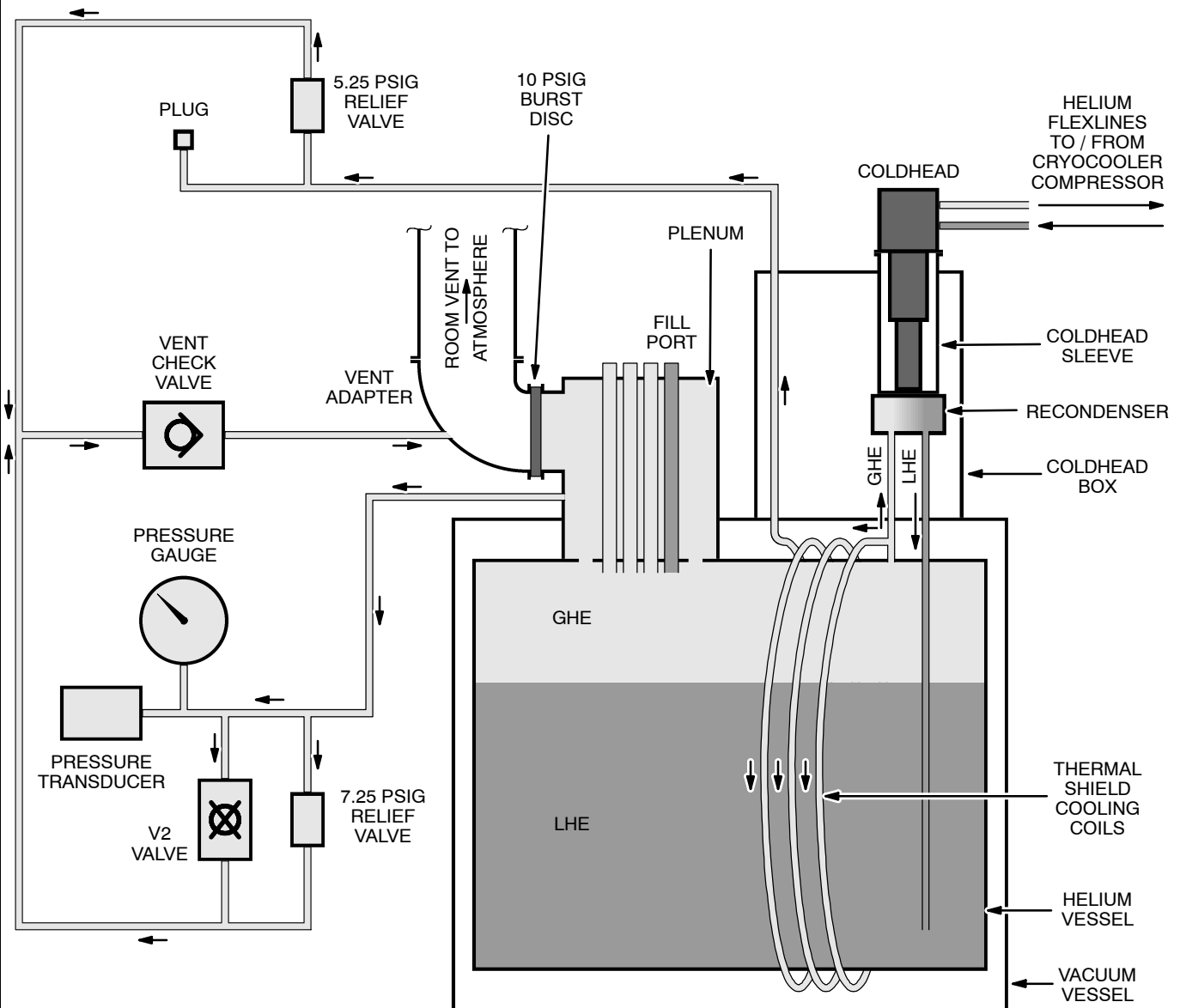
2-2-3 Coldhead Replacement Tools / Equipment (continued)

- Coldhead Replacement Gasket Kit (2171620)
- Getter Bag (2290272; includes 12 bags, 1 per Coldhead change)
- Universal Coldhead Replacement Tool Kit (2300306)
- Nonmagnetic Coldhead Wrench Kit (46-294804G1)
- For fixed site magnets:
 - Coldhead Insertion and Purge Cover Tools Kit (2232674), which includes:
 - Fixed Site Coldhead Insertion Fixture (2228876)
 - Purge Cover Plate Assembly (2232560)
 - Heat Exchanger Cover Plate (2275776) (not included in kit 2232674)
- For mobile site magnets:
 - LCC Mobile Coldhead Insertion / Removal Tool Kit (2275779), which includes:
 - LCC Mobile Site Coldhead Insertion Fixture (2275599)
 - Mobile Site Coldhead Lifting Tool (2258802)
 - Heat Exchanger Cover Plate (2275776)
- Coldhead Stand (Holding Tool) (2214919)
- Portable RuO Temperature Monitor (2171219; included in RuO Temperature Monitor Kit 2183710)
- Pump-Out Port Dual O-Ring Plug (2236606)
- Portable O₂ Monitor (2106236 or 2106237)
- Full, nonmagnetic cylinder(s) of nitrogen gas
- Cylinder(s) of dry nitrogen gas
- Gas Regulator and Hose Kit (46-306734G1)
- Aluminum Tape (46-260802P1 / P6) or Permacel Tape
- Flashlight
- Scotch-Brite™ pads
- Leather gloves (for opening / closing motor shield and handling removed coldhead)
- Vacuum with nonmagnetic extension hose

2-3 TROUBLESHOOTING GUIDE

Note

Illustration 2-3 shows the path of liquid helium (LHe) and gaseous helium (GHe) through the helium vessel and plumbing of an installed magnet.



HELIUM PATH SCHEMATIC
ILLUSTRATION 2-3

2-3 TROUBLESHOOTING GUIDE (continued)

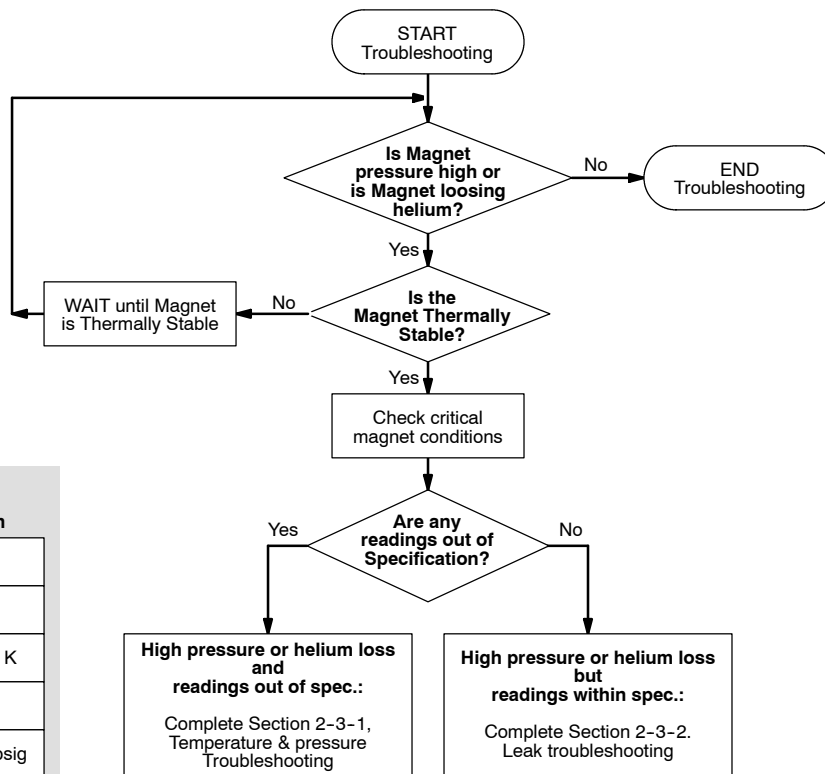
IMPORTANT !!!

Follow the temperature and pressure troubleshooting procedures in sequence to minimize unnecessary Coldhead removal / replacement. Repair all problems as discovered, then proceed with additional troubleshooting, as required.

Note
High helium pressure, helium loss or high Coldhead temperatures occurring after an installation, magnet ramping / shimming, helium fill, power outage or Coldhead replacement
DO NOT NECESSARILY INDICATE
a problem with magnet thermal performance.

CRITICAL MAGNET CONDITIONS	
	Specification
Second Stage Coldhead Temp. (Use Portable RuO Monitor 2171219)	< 4.4 K
First Stage Coldhead Temp. (Magnet Monitor)	< 45 K
Recondenser Temperature (Calculate Recon. T. - 2nd Stage T.)	> 3.9 but < 4.5 K
Delta T (Use Portable RuO Monitor 2171219)	< 0.2 K
Magnet Helium Pressure (Magnet Monitor)	> 0.3 but < 4.7 psig

Note
To determine if the helium pressure is constant with the Recondenser temperature refer to Illustrations 2-2 and 2-3.



INITIAL TROUBLESHOOTING
ILLUSTRATION 2-1

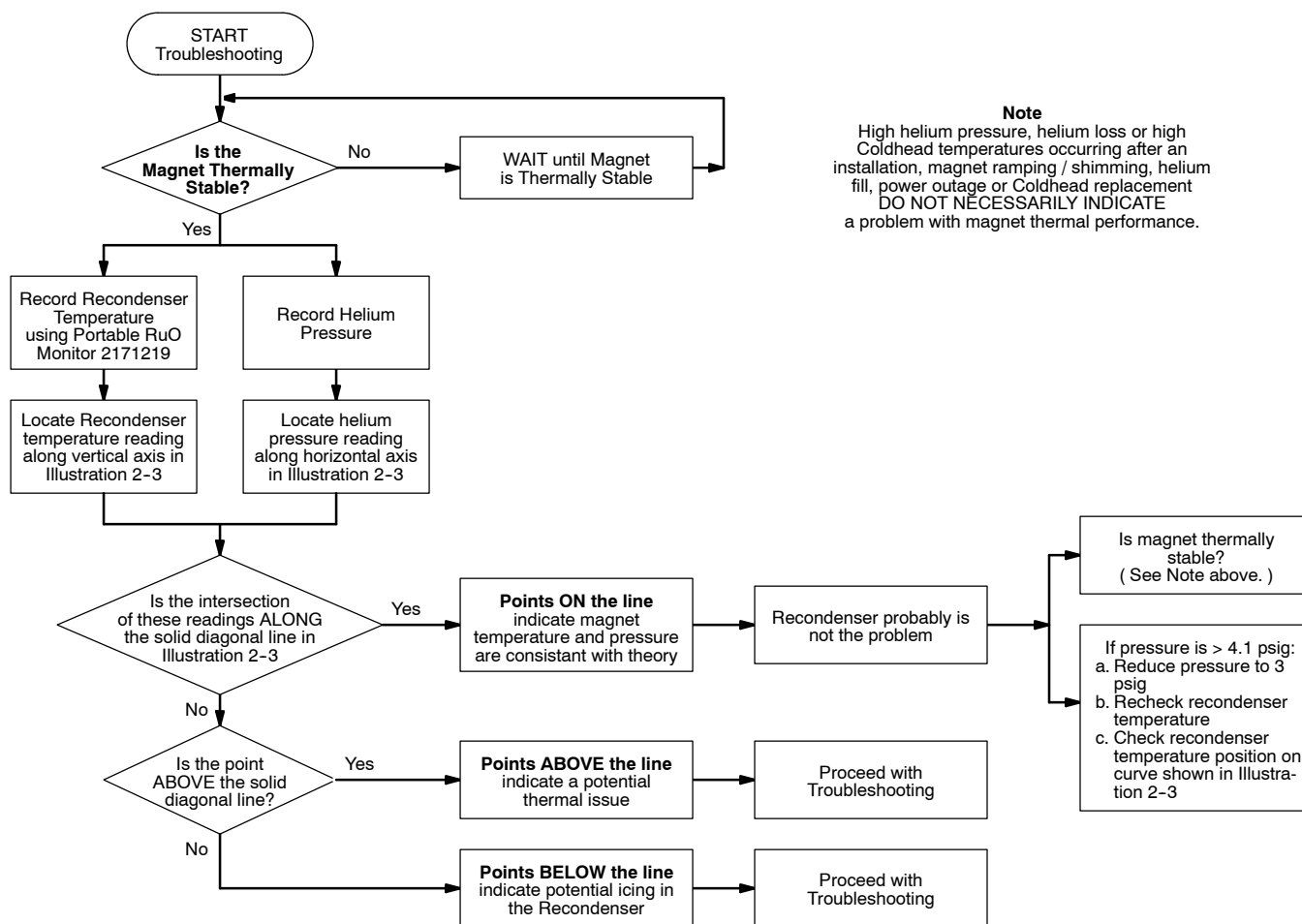
IMPORTANT !!!

Second-Stage Coldhead and Recondenser readings MUST be taken using the Portable RuO Temperature Monitor (2171219) connected directly to the RuO outputs on the Coldhead and Sleeve.

Note

The Portable RuO Temperature Monitor (2171219) readings of the Second-Stage Coldhead may show a small oscillation (0.1 – 0.25 K) resulting from Displacer motion inside the Coldhead Sleeve and the monitor's sampling rate. Record the average temperature reading: $(T_{\min} + T_{\max}) / 2$.

2-3 TROUBLESHOOTING GUIDE (continued)

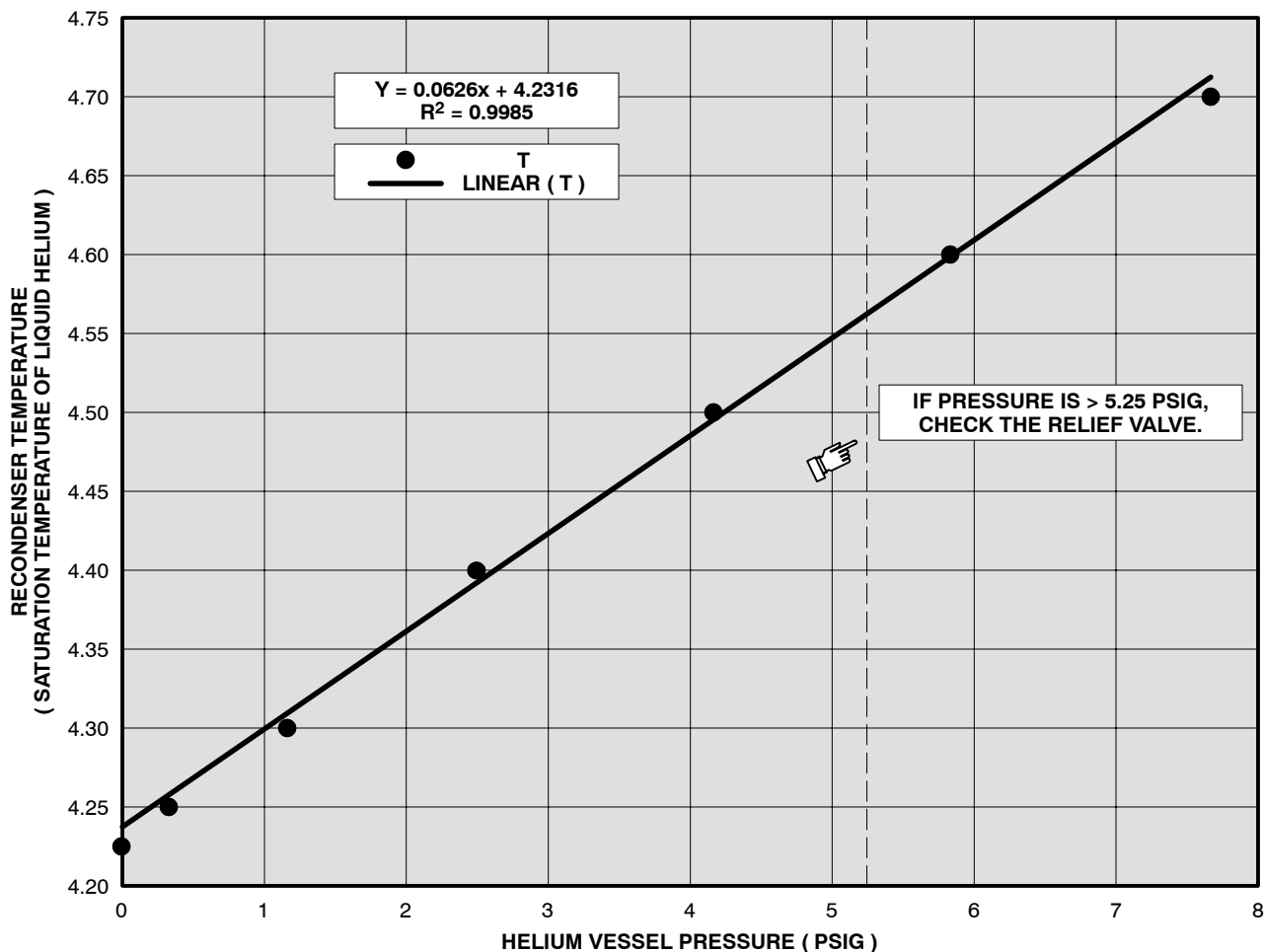


INITIAL TROUBLESHOOTING FLOW CHART
ILLUSTRATION 2-2

Note

Use the flow chart in Illustration 2-2 in conjunction with the graph in Illustration 2-3 to determine if vapor pressure is consistent with RuO readings.

2-3 TROUBLESHOOTING GUIDE (continued)



LIQUID HELIUM (LHE) SATURATION TEMPERATURE
(APPROXIMATE RECONDENSER TEMPERATURE VS. HELIUM VESSEL PRESSURE)
ILLUSTRATION 2-3

Note

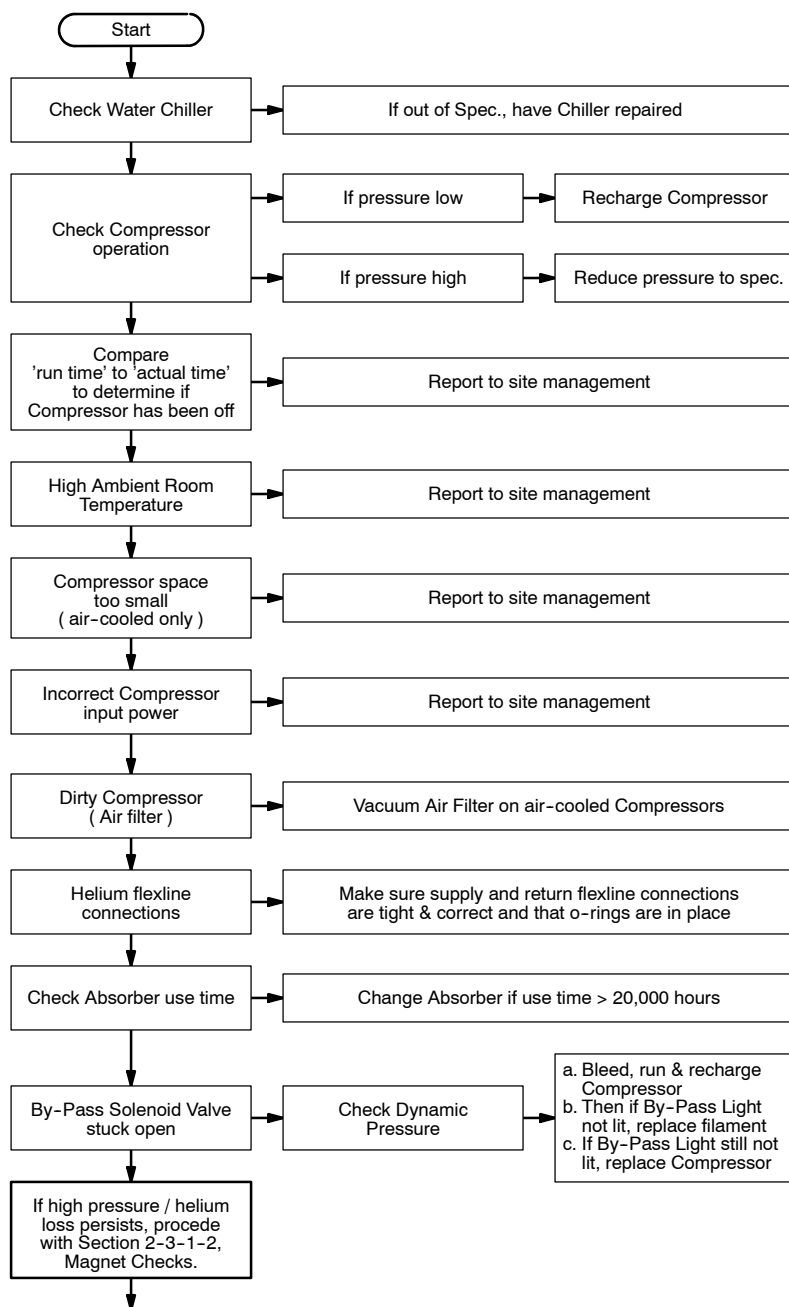
Use the graph in Illustration 2-3 in conjunction with the flow chart in Illustration 2-2 to determine if vapor pressure is consistent with RuO readings.

WARNING!

THE CRYOSTAT EXHAUST SYSTEM IS DESIGNED WITH A 5.25 PSIG RELIEF VALVE, ALLOWING INTERNAL CRYOSTAT PRESSURE TO BUILD UP TO 5.25 PSIG. OBSERVE THE INTERNAL CRYOSTAT PRESSURE GAUGE READING, SET MAGNET MONITOR PRESSURE TO 0.5 PSIG AND VENT THE CRYOSTAT DOWN TO 0.3 PSIG THROUGH VENT VALVE V2 BEFORE REMOVING EITHER THE RAMP OR FILL CAPS OR LOOSENING ANY COMPONENT RESULTING IN THE RELEASE OF CRYOSTAT PRESSURE. THIS WILL PREVENT EXPOSURE TO RAPID EXHAUSTING OF COLD HELIUM GAS DURING A SERVICE OPERATION.

2-3-1 Temperature / Pressure Troubleshooting**2-3-1-1 Chiller and Compressor Checks****Note**

Also refer to SHI SRDK Cryocooler Service Manual CO3722-055C.

**Reference Information**

Water flow and temperature minimums:
Refer to SHI Compressor manual, Section 1-1, Tables 1-1, 1-2 and Figure 1-1

Compressor Pressures
Static: 16.3 - 16.8 Kgf/cm²G (1.6 - 1.65 MPa)
Dynamic: 21.4 - 23.5 Kgf/cm²G (2.1 - 2.3 MPa)

Temperature < 28C (82F)

Refer to SHI manual CVSA-71A, page 18

Refer to SHI Compressor manual

Refer to SHI manual CVSA-71A, page 23

Refer to SHI manual CVSA-71A

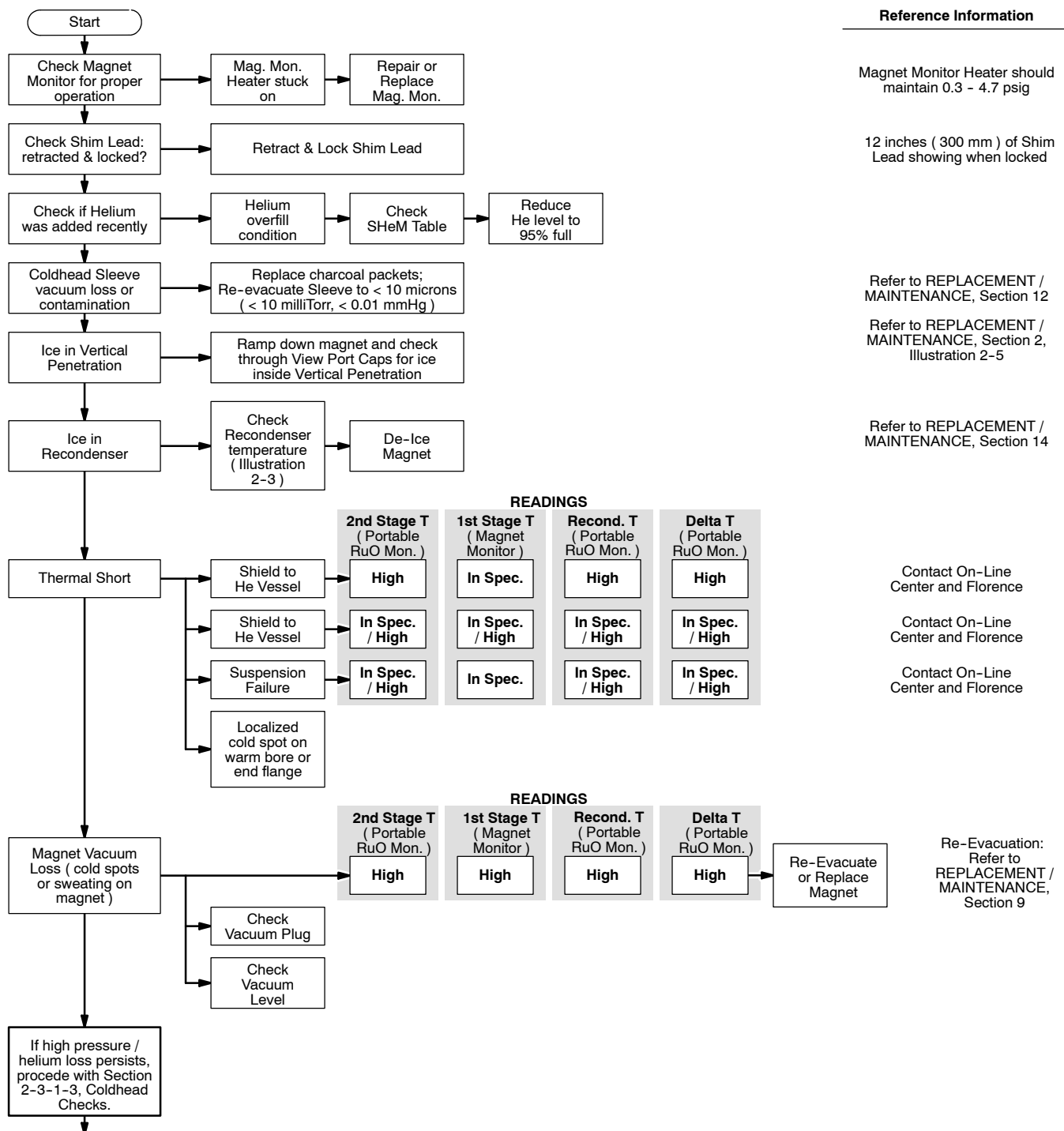
Refer to SHI manual CVSA-71A, page 23

By-Pass Light lit for normal operation.

Dynamic Compressor pressure:
21.4 - 23.5 Kgf/cm²G (2.1 - 2.3 MPa)

**HIGH PRESSURE / TEMPERATURE TROUBLESHOOTING FLOW CHART:
SITE PROBLEMS AND CHILLER / COMPRESSOR CHECKS**

ILLUSTRATION 2-4

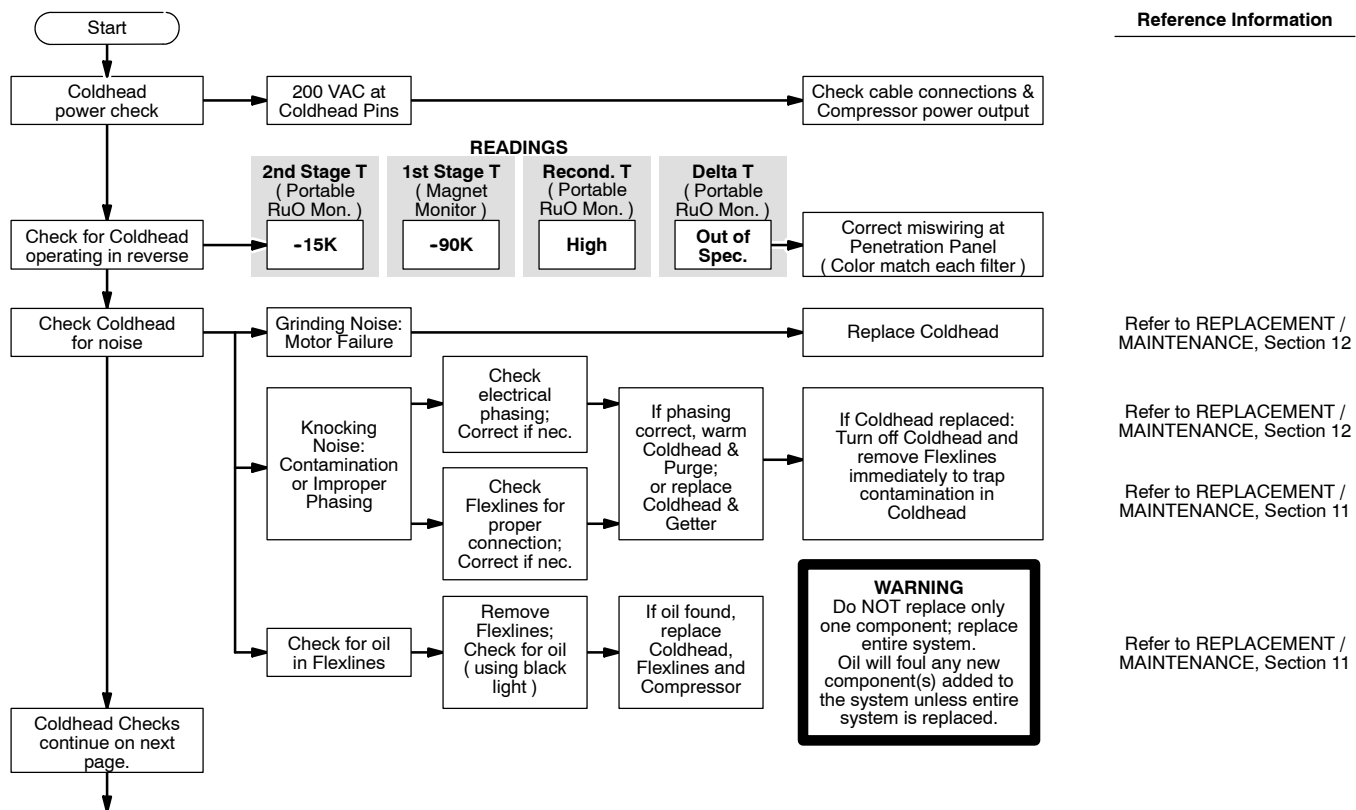
2-3-1 Temperature / Pressure Troubleshooting (continued)**2-3-1-2 Magnet Checks**

**HIGH PRESSURE / TEMPERATURE TROUBLESHOOTING FLOW CHART:
MAGNET AND COLDHEAD INTERFACE CHECKS**

ILLUSTRATION 2-5

2-3-1 Temperature / Pressure Troubleshooting (continued)**2-3-1-3 Coldhead Checks**

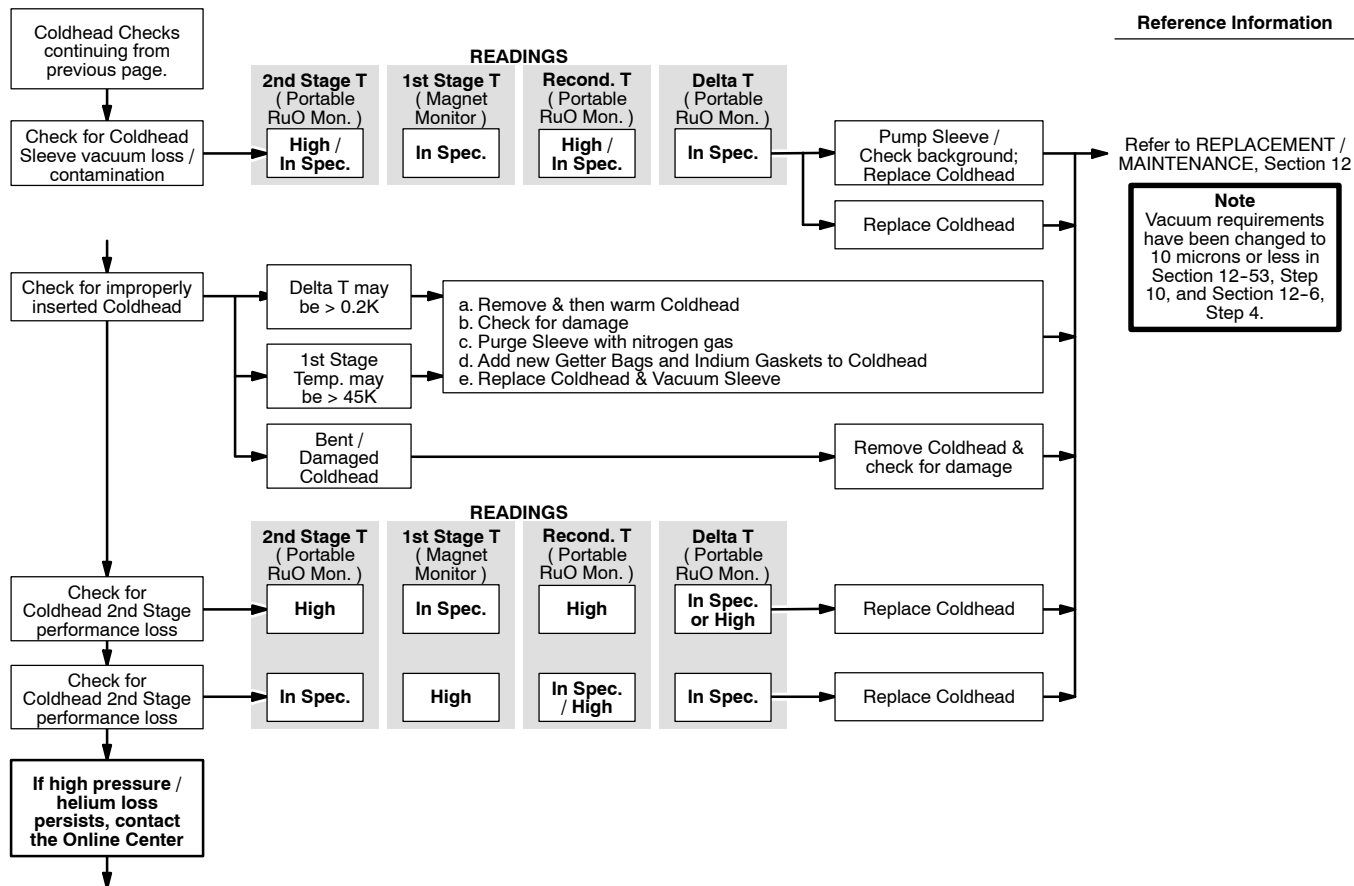
Only perform Coldhead Checks after completion of Chiller and Compressor Checks.



**HIGH PRESSURE / TEMPERATURE TROUBLESHOOTING FLOW CHART:
COLDHEAD CHECKS
ILLUSTRATION 2-6**

IMPORTANT !!!

Use a black light or a clean, lint-free white cloth to check for oil contamination whenever Flexlines are disconnected. If oil contamination is found, the Coldhead, Flexlines and Compressor must be replaced.

2-3-1 Temperature / Pressure Troubleshooting (continued)**2-3-1-3 Coldhead Checks (continued)**

**HIGH PRESSURE / TEMPERATURE TROUBLESHOOTING FLOW CHART:
COLDHEAD CHECKS**
ILLUSTRATION 2-6 (CONTINUED)

IMPORTANT !!!

Use a black light or a clean, lint-free white cloth to check for oil contamination whenever Flex-lines are disconnected. If oil contamination is found, the Coldhead, Flexlines and Compressor must be replaced.

2-3-2 Leak Troubleshooting

2-3-2-1 Introduction

Operation of the magnet in the “C0BO” (Controlled Zero Boil-Off) mode with internal helium pressure > 3.9 psig increases the potential of helium leaks. A number of small helium leaks will have a significant effect on boil-off and result in the measurable loss of helium over time.

Ice or condensation on plenum plumbing indicates a leak in the helium plumbing.

Major helium leaks will depressurize the helium vessel and can result in cryopumping and icing inside the vessel.

WARNING!

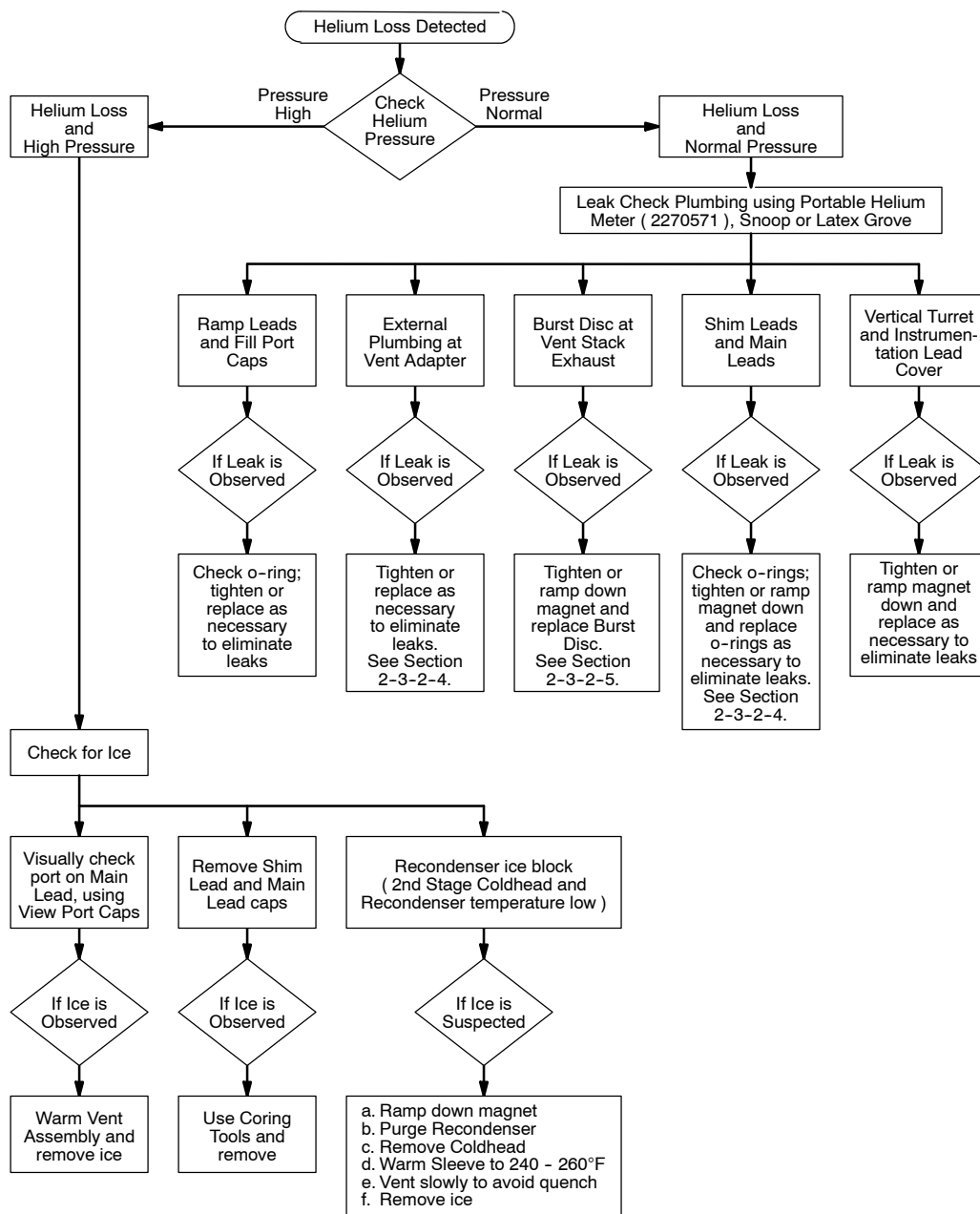
MAKE SURE THE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON — OR THE HATCH IS OPENED IF A MOBILE VAN — BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, INCLUDED WITH THIS MANUAL.

WEAR PROTECTIVE CLOTHING, NONABSORBENT GLOVES AND GOGGLES WHEN REPLACING THE BURST DISC ON A COLD VENT SYSTEM.

2-3-2 Leak Troubleshooting (continued)**Reference Information**

If not in "C0BO" Mode,
turn off Cryocooler and allow
pressure in Helium Vessel to
build to 3.9 psig

Refer to REPLACEMENT /
MAINTENANCE, Section 2



LEAK TROUBLESHOOTING FLOW CHART
ILLUSTRATION 2-7

2-3-2 Leak Troubleshooting (continued)**2-3-2-2 Pressure**

1. Make sure helium pressure is < 3.9 psig. Check Magnet Monitor operation if pressure is low.

2-3-2-3 Leak Detection

Commercial leak detector fluids or instruments available in the HVAC industry can be used to determine small leaks in the helium plumbing and venting. Some examples available through HVAC distributors are:

- Bird Dog Quart Spray (p/n DOG-1Q)
- Big Blue (p/n Rt100S / Rt100G)
- Snoop
- Portable Leak Detector

1. When the pressure is < 3.9 psig, check the following areas for leaks with a leak detector fluid or instrument.
 - All external plumbing joints
 - Vertical Turret and Instrumentation Lead Cover
 - Main Lead and fill port caps
 - Shim Lead collar / cap
 - Pressure Gauge
 - Shim Lead Pins

2. Repair any leaks found and then test the repairs.

2-3-2-4 Plumbing and O-Ring Leaks**Note**

Replacement o-rings and a Burst Disc are contained in the Field Spare Parts Kit (46-294744G5).
See RENEWAL PARTS, Section 3.

1. Remove the outlet plumbing (Vent Adapter side) for the 5.25 psig relief valve and the 7.25 psig bypass valve.
2. Remove the cap from the Nupro valve at the top of the Shim Lead.

Note

Suitable alternative leak checking equipment can be used in Steps 3 and 4.

3. Attach a small balloon tightly to the exhaust side of the 5.25 psig relief valve, 7.25 psig bypass valve and the Nupro valve.
4. Observe any expansion of the balloon from leaking helium gas.
5. Replace any valve that has suspected leaks and retest the new valve for leaks. Valve part numbers shown in RENEWAL PARTS, Section 2.

2-3-2 Leak Troubleshooting (continued)**2-3-2-5 Leaks at Burst Disc**

1. Ramp the magnet down to zero field in conformance with REPLACEMENT / MAINTENANCE, Section 3.
2. Remove the Vent Adapter.
3. Check the Burst Disc for cracks or significant leaks around the gasket. Replace any cracked or leaking Burst Disc. See REPLACEMENT / MAINTENANCE, Section 5.



Small (micro) bubbles may appear between the gasket and flange. This is normal for this gasket material. Make sure the Belleville washers are flattened and that no major leaks (continuous bubbles over 0.08 inch or 2 mm in diameter) are present. Foam-like bubbles may be present.

2-4 ASSISTANCE FOR FLORENCE-MANUFACTURED MAGNETS

When the furthest right digit on the Magnet Monitor pressure gauge decreases by a few units over 1 - 2 hours there is a 90% probability that high pressure / helium loss issue(s) are resolved. Magnet pressure usually returns to and stabilizes at ~ 4 psig in less than one week.

If the root cause of an issue is not found after completing Section 2-3, Troubleshooting Guide, collect the information in Table 2-2 and the online center for assistance.

TABLE 2-2
MAGNET INFORMATION
(COLLECT BEFORE CALLING FOR ASSISTANCE)

SYSTEM ID. _____
MAGNET NUMBER _____
MAGNET INSTALLATION DATE _____

SITE TYPE (CIRCLE ONE): _____ MOBILE / FIXED
MAC TEAM LEADER _____

TWO-STAGE CRYOCOOLER COMPRESSOR	SERIAL NUMBER	
	INSTALL DATE	
	RUN TIME	
	STATIC PRESSURE	
	DYNAMIC PRESSURE	
	INLET WATER FLOW	
	WATER TEMPERATURE AT INLET	
TWO-STAGE CRYOCOOLER COLDHEAD	SERIAL NUMBER	
	SECOND STAGE (HAND-HELD RUO)	
	RECONDENSER TEMPERATURE (HAND-HELD RUO)	
	DELTA T	
	FIRST STAGE TEMPERATURE	
	SLEEVE VACUUM	
	NUMBER OF GETTER BAGS	
	DATE GETTER BAGS LAST CHANGED	
	NOISE	
MAGNET PRESSURE		
HELIUM LEVEL, CYLINDRICAL		
FROST ON PLUMBING		
MAGNET MONITOR HEATER OPERATION VERIFICATION		
COMMENTS		